Local water and new development

How could a new building affect local water supplies and rivers?

Our current tool estimates water used to cool equipment.

For town planners, researchers, and water providers.

Current test data: generated examples.
Next: test measured water use at actual sites.

How the program works

1. Inputs

Six numeric features
Method + location

2. Estimate

XGBoost learned trees
Daily water use (MGD)

3. Check

Symbolic energy rules
Constrain + flag (E3)

4. Result

Gallons/day + range
Warnings + sources

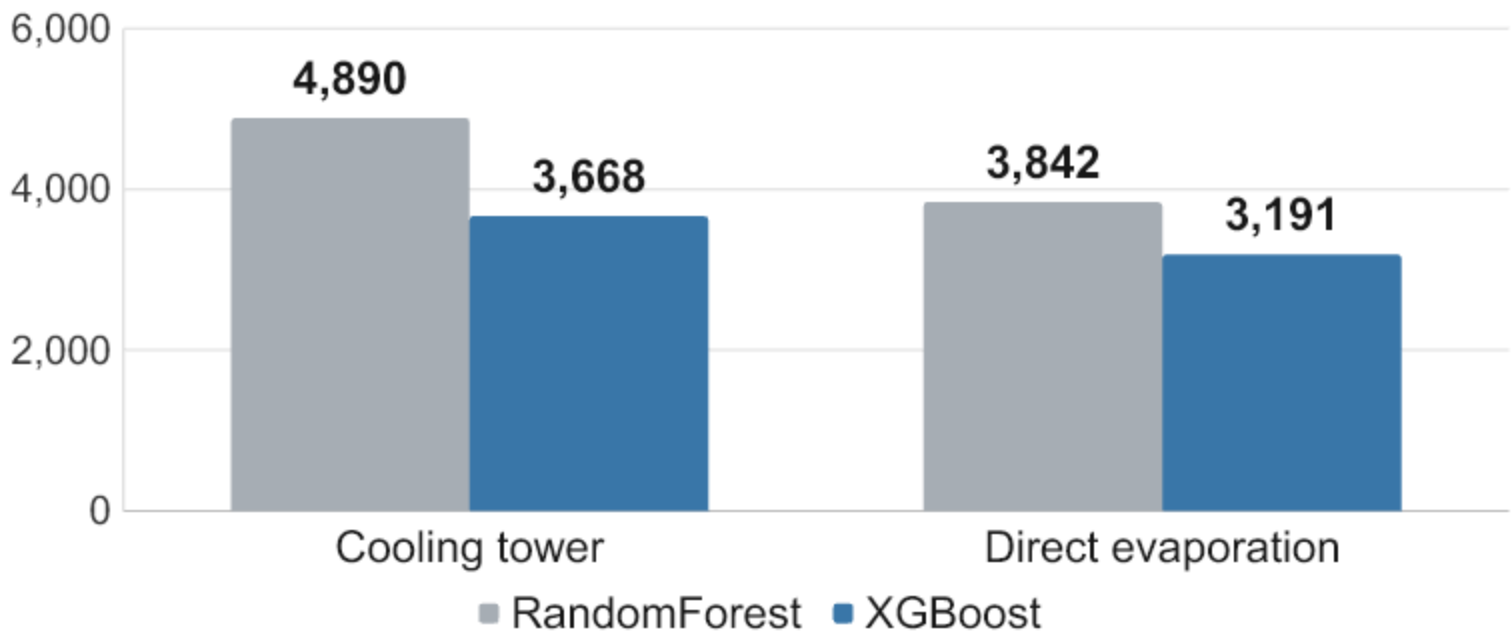
Features: MW, air/wet-bulb temperature, river flow, PDSI drought, month.

Permit path: text -> TF-IDF vectors + HUC/location filter -> exact quotes (E4).

E3: -1 MGD -> 0 + warning. E4: wrong watershed -> no permit passage.

Test 1: Which model makes smaller errors?

Average error in gallons per day. Lower is better.



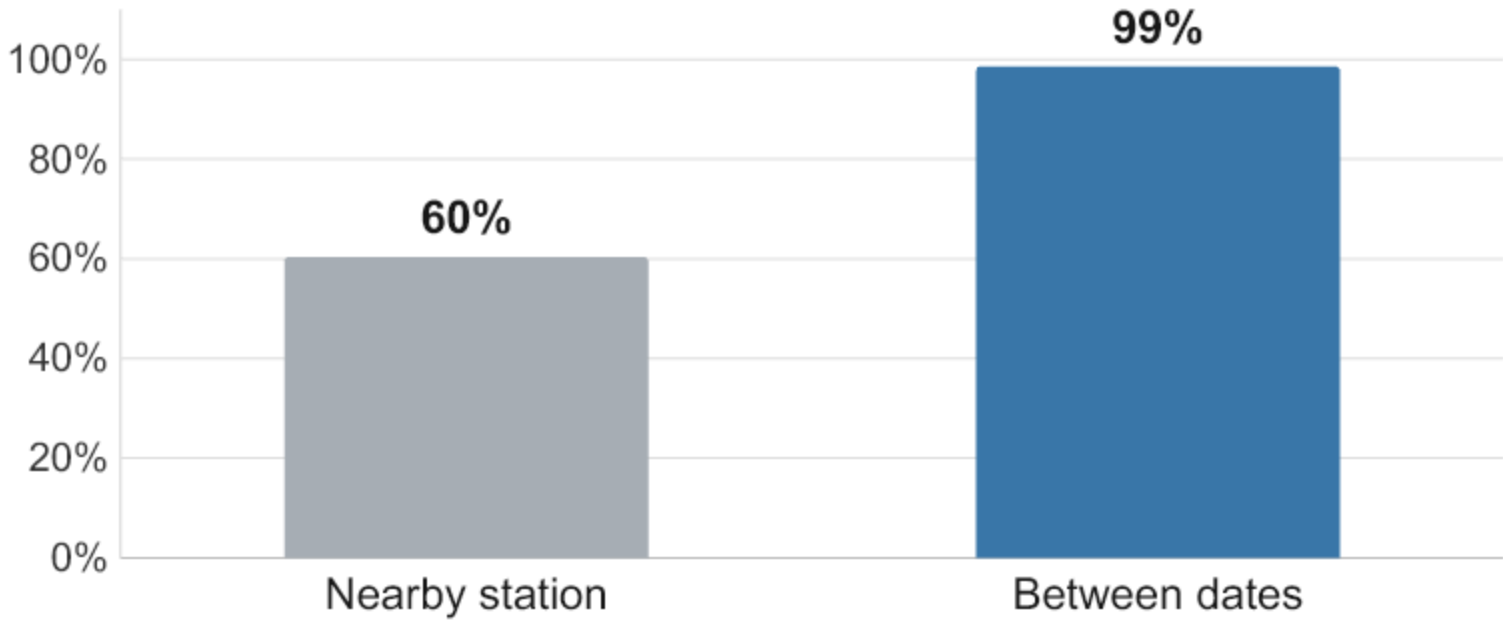
Both models learn from examples.

XGBoost made smaller errors in this test.

E1 / regression-comparison.csv / 333 test examples per cooling method

Test 2: What if drought readings are missing?

Bad nearby readings: share of changes within the allowed limit



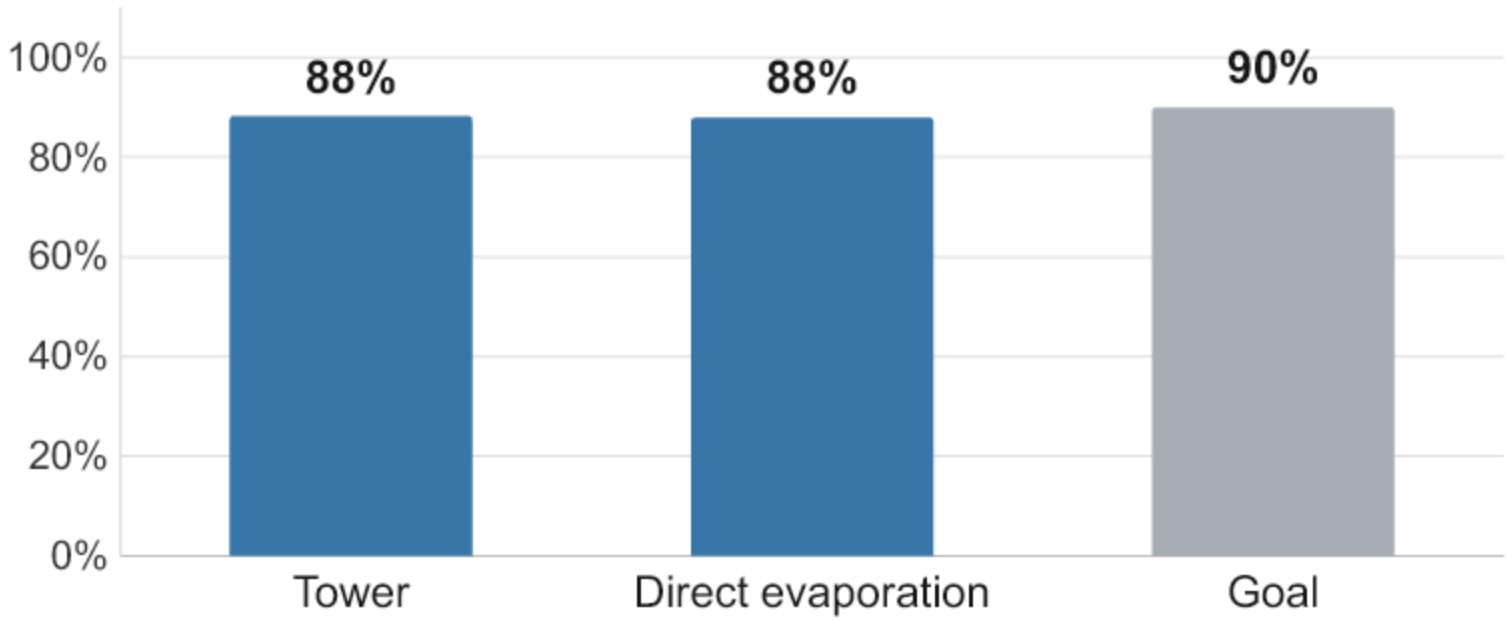
Within 50,000 gallons/day of the no-gap result.

Nearby readings need to represent the same conditions.

Between-date filling needs later readings; use it for historical gaps.

Test 3: Did the prediction range work?

Share of test answers inside the low-to-high range



Goal: 90 out of 100.
Got: about 88.

Use separate data
to choose the model
and set its range.

E5 / interval-coverage.csv / Tower: 294 of 333 / Direct evaporation: 293 of 333

What we will keep and improve

Keep (Adopt)	XGBoost (E1), rules (E3), local quotes (E4).
Improve (Modify)	Donor checks (E2); separate calibration (E5).
Reject	Fit as accuracy: strong fit, missed coverage (E1/E5).
Wait (Defer)	Neural/LLM methods: no fair measured-data test yet.

Next: train on some sites and test on different sites.